

Internship Week 1 Project: House Price Prediction

Summary Report.

Task 5 — Insights & Summary

Based on our machine learning pipeline, **property size (area)**, the **number of bathrooms**, and whether the house features **air conditioning** are the variables that influence house prices the most.

In plain terms, both predictive models are moderately accurate, capturing roughly 61% to 65% of the variances in pricing data (with Linear Regression slightly outperforming the Random Forest model on test predictions with an R^2 score of `0.6529` vs `0.6104`).

An interesting surprise in the dataset is that having a basement or more bedrooms does not increase properties' valuation nearly as significantly as adding a finished bathroom or central air cooling units.

Business Recommendation: For real estate businesses looking to maximize house resale flip margins, investment capital should be prioritized toward adding modern air conditioning infrastructure and adding/renovating luxury bathroom units over simply expanding raw bedroom count.

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